

SAN JUAN MARIN INTL, Puerto Rico

WMO: 785260

Lat: 18.4317N

Lon: 65.9919W

Elev: 4

StdP: 101.28

Time Zone: -4.00 (W04)

Period: 94-19

WBAN: 11641

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	21.1	21.7	15.3	10.9	24.7	16.3	11.6	24.9	10.0	26.7	9.4	26.9	0.9	150	0.393

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.6	33.1	25.3	32.1	25.5	31.6	25.5	27.2	30.2	26.9	30.0	26.6	29.9	5.3	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.3	21.7	28.7	25.9	21.3	28.6	25.6	20.8	28.5	85.9	30.4	84.1	30.1	82.8	29.9	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.7	8.1	19.9	34.3	1.4	0.7	18.9	34.8	18.1	35.3	17.3	35.7	16.3	36.2		
(5)			17.1	27.9	1.2	0.7	16.3	28.4	15.6	28.8	15.0	29.3	14.1	29.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.1	25.3	25.3	25.8	26.6	27.4	28.3	28.4	28.5	28.5	28.1	26.9	26.0
	DBStd	1.56	0.94	0.96	1.03	1.08	1.11	0.94	0.88	0.87	0.99	0.99	0.96	0.97
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6239	474	429	489	497	539	549	569	573	556	560	507	495
	CDD18.3	3197	216	196	231	247	281	299	311	315	306	302	257	237
	CDH23.3	30712	1406	1277	1696	2163	2774	3323	3528	3613	3486	3228	2374	1844
	CDH26.7	8983	213	220	364	549	786	1137	1229	1292	1258	1043	564	329
Wind	WSAvg	3.5	3.4	3.7	3.7	3.8	3.6	3.9	4.3	3.7	3.0	2.6	2.9	3.4
Precipitation	PrecAvg	1411	82	57	55	105	145	108	128	150	147	138	167	121
	PrecMax	2275	281	170	235	381	381	345	360	472	377	383	448	427
	PrecMin	829	15	5	0	0	0	4	28	47	45	28	10	15
	PrecStd	335	52	34	41	80	94	74	63	71	74	73	89	70
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.4	30.9	31.9	32.9	33.2	33.5	33.1	33.3	33.9	33.4	31.6	30.9
		MCWB	23.0	22.8	22.7	24.2	24.7	25.0	25.5	25.6	25.6	25.3	25.1	23.9
	2%	DB	29.4	29.6	30.4	31.2	31.7	32.2	31.9	32.1	32.8	32.0	30.7	29.8
		MCWB	23.1	22.8	23.3	24.0	24.8	25.4	25.8	26.0	25.7	25.7	25.2	23.9
	5%	DB	28.7	28.9	29.5	30.2	30.9	31.5	31.3	31.4	31.7	31.2	30.1	29.2
		MCWB	23.2	22.9	23.1	24.0	25.0	25.5	25.7	26.0	25.9	25.6	25.0	24.0
	10%	DB	28.1	28.2	28.7	29.4	30.1	30.9	30.9	31.0	31.0	30.6	29.4	28.6
		MCWB	23.1	22.8	23.0	23.8	24.9	25.5	25.7	25.9	25.8	25.6	24.9	23.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.1	24.7	25.1	26.0	26.9	27.4	27.5	27.6	27.6	27.6	26.7	26.0
		MCDB	27.8	28.0	28.5	29.4	29.7	29.9	30.3	30.6	30.5	30.0	29.4	28.5
	2%	WB	24.4	24.1	24.4	25.2	26.3	26.7	26.9	27.1	27.0	26.8	26.2	25.2
		MCDB	27.3	27.6	28.0	28.8	29.4	29.9	30.0	30.3	30.5	29.8	29.1	28.0
	5%	WB	24.0	23.6	24.0	24.7	25.8	26.3	26.5	26.7	26.6	26.4	25.7	24.7
		MCDB	27.0	27.1	27.5	28.4	29.0	29.8	29.8	30.0	30.3	29.7	28.6	27.6
	10%	WB	23.6	23.2	23.5	24.3	25.3	25.9	26.2	26.3	26.2	26.0	25.2	24.3
		MCDB	26.6	26.6	27.1	28.0	28.5	29.6	29.7	29.8	30.0	29.5	28.2	27.2
Mean Daily Temperature Range	5% DB	MDBR	5.7	6.0	6.2	6.1	5.8	5.8	5.6	5.6	5.8	5.9	5.5	5.4
		MCDBR	6.6	7.0	7.4	7.2	6.7	6.4	5.9	6.0	6.4	6.4	6.3	6.2
		MCWBR	2.7	2.7	2.7	2.7	2.5	2.5	2.5	2.5	2.6	2.6	2.6	2.6
	5% WB	MCDBR	5.9	6.3	6.5	6.3	5.9	5.9	5.6	5.7	5.9	6.0	5.8	5.4
		MCWBR	2.9	2.9	2.8	2.8	2.7	2.7	2.7	2.8	2.8	2.8	2.7	2.7
Clear-Sky Solar Irradiance	taub	0.372	0.387	0.387	0.414	0.455	0.529	0.528	0.491	0.444	0.393	0.387	0.378	
	taud	2.548	2.494	2.491	2.445	2.340	2.093	2.094	2.230	2.391	2.552	2.566	2.561	
	Ebn at Noon	912	914	923	894	846	778	781	816	857	898	891	894	
	Edh at Noon	97	107	111	117	129	163	163	143	121	100	95	94	
All-Sky Solar Radiation	RadAvg	4.54	5.28	5.86	6.16	5.93	6.04	6.12	5.96	5.63	5.03	4.41	4.24	
	RadStd	0.16	0.12	0.24	0.34	0.41	0.35	0.31	0.20	0.36	0.33	0.32	0.19	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.33	+0.59	N/A	N/A	N/A	N/A	N/A	+113	+112
(47) Regional (8 neighbors)		+0.28	+0.47	N/A	+0.33	N/A	N/A	N/A	+113	+112

Nomenclature: See separate page